Penetration Testing CY321 Lab Client Machines: Windows Server, Windows 10, Kali Linux, and Metasploitable 2 Setup

Overview

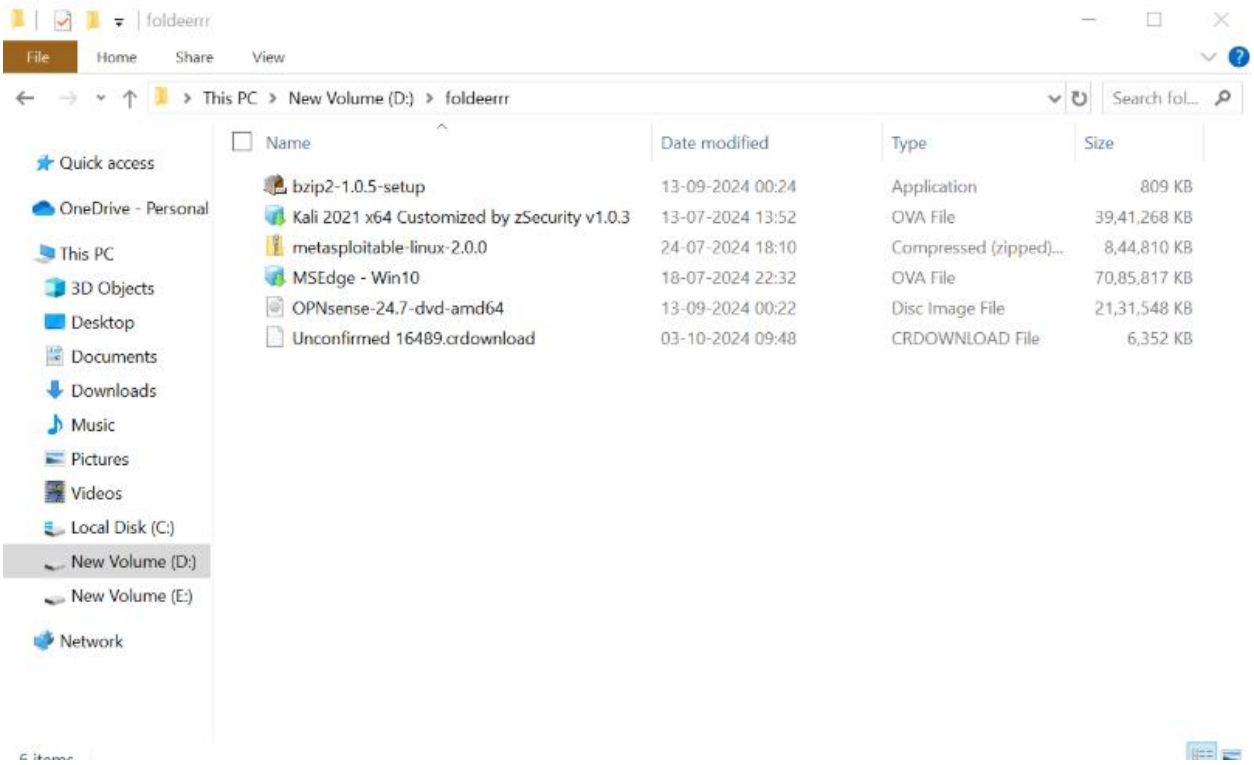
In this, I am going to set up the following client machines and servers in my Homelab:

- **Windows Server 2019**: Configured as a domain controller with **Active Directory**.
- **Windows 10 Client**: Joined to the domain to simulate a corporate environment.
- **Kali Linux**: Used for penetration testing and vulnerability scanning i.e to perform different attacks on windows machine and metasploitable machine.
- **Metasploitable 2**: A vulnerable target machine used for performing attacks and exploiting its vulnerabilities.

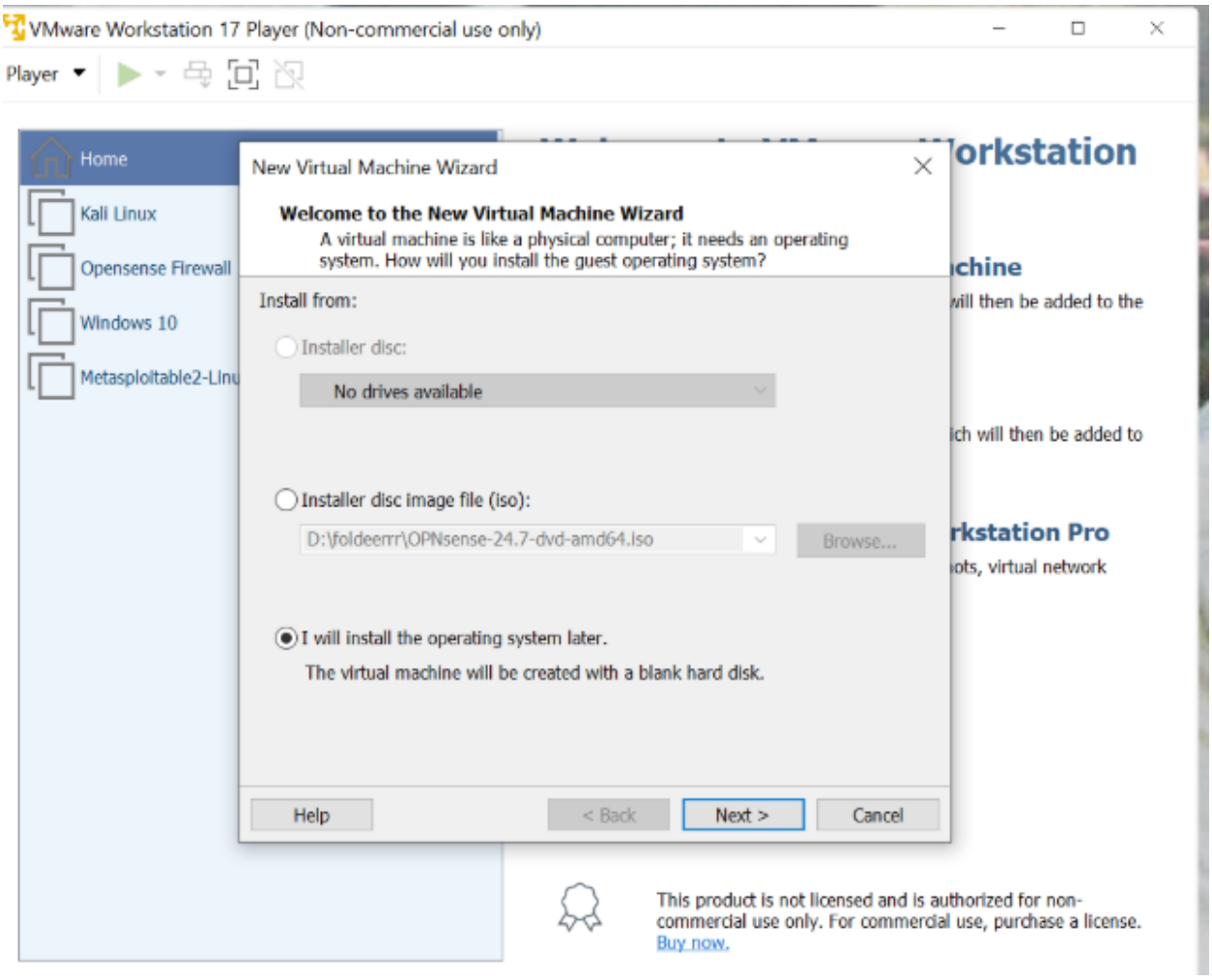
Step 1: Set Up the Windows Server 2019 (Active Directory Domain Controller)

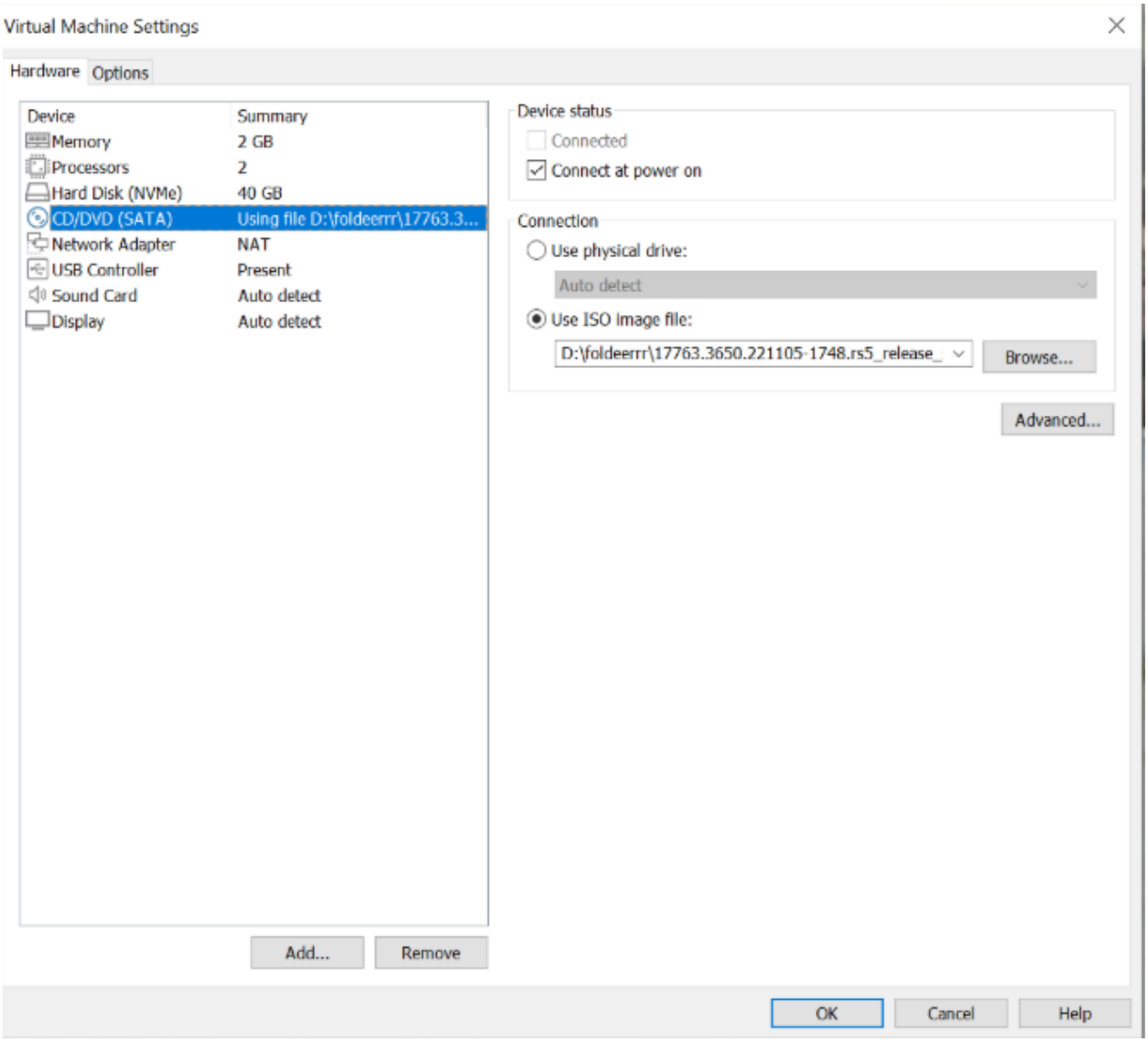
1.1 Install Windows Server on VMware

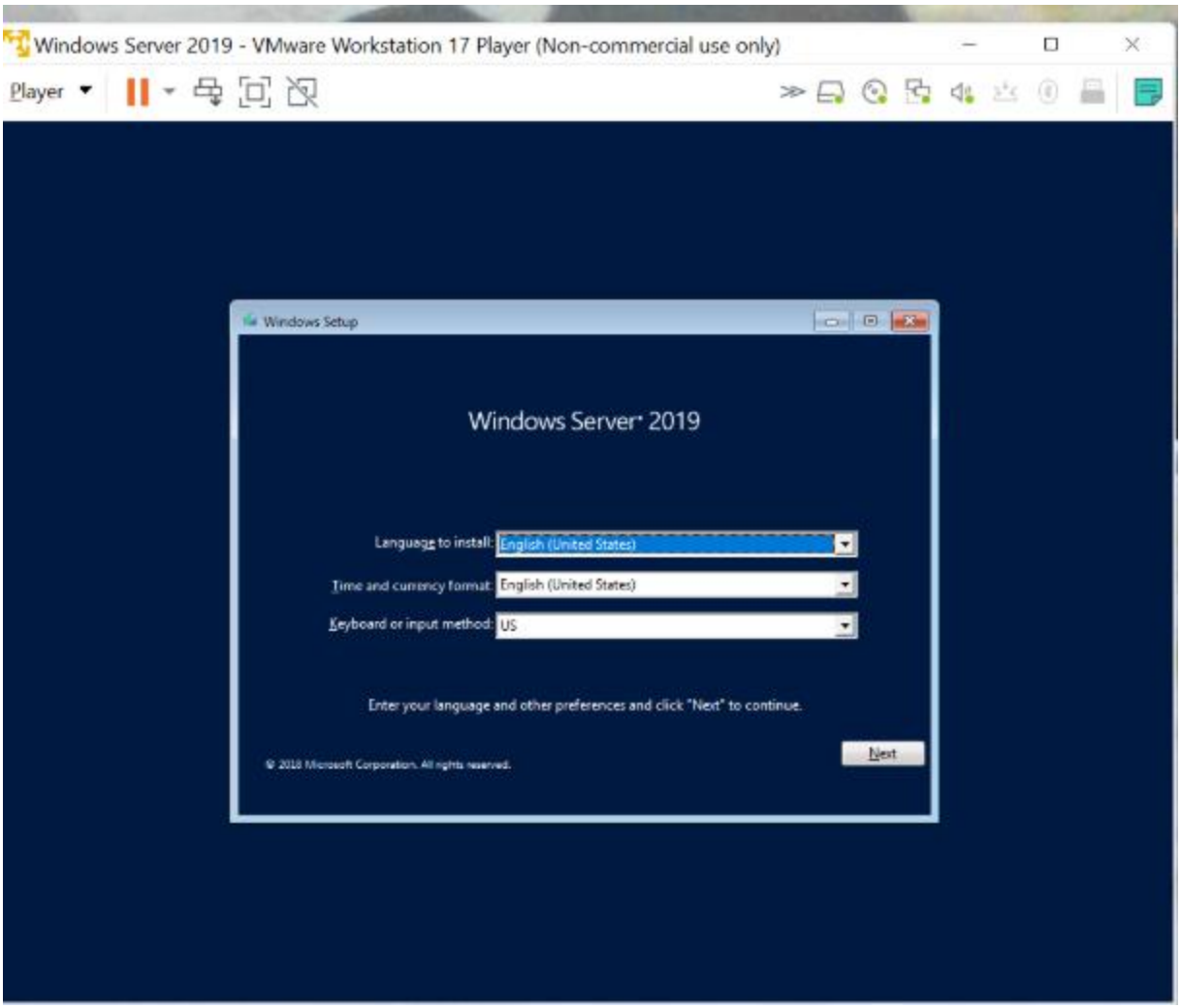
1. **Download Windows Server ISO** from Microsoft's official site.

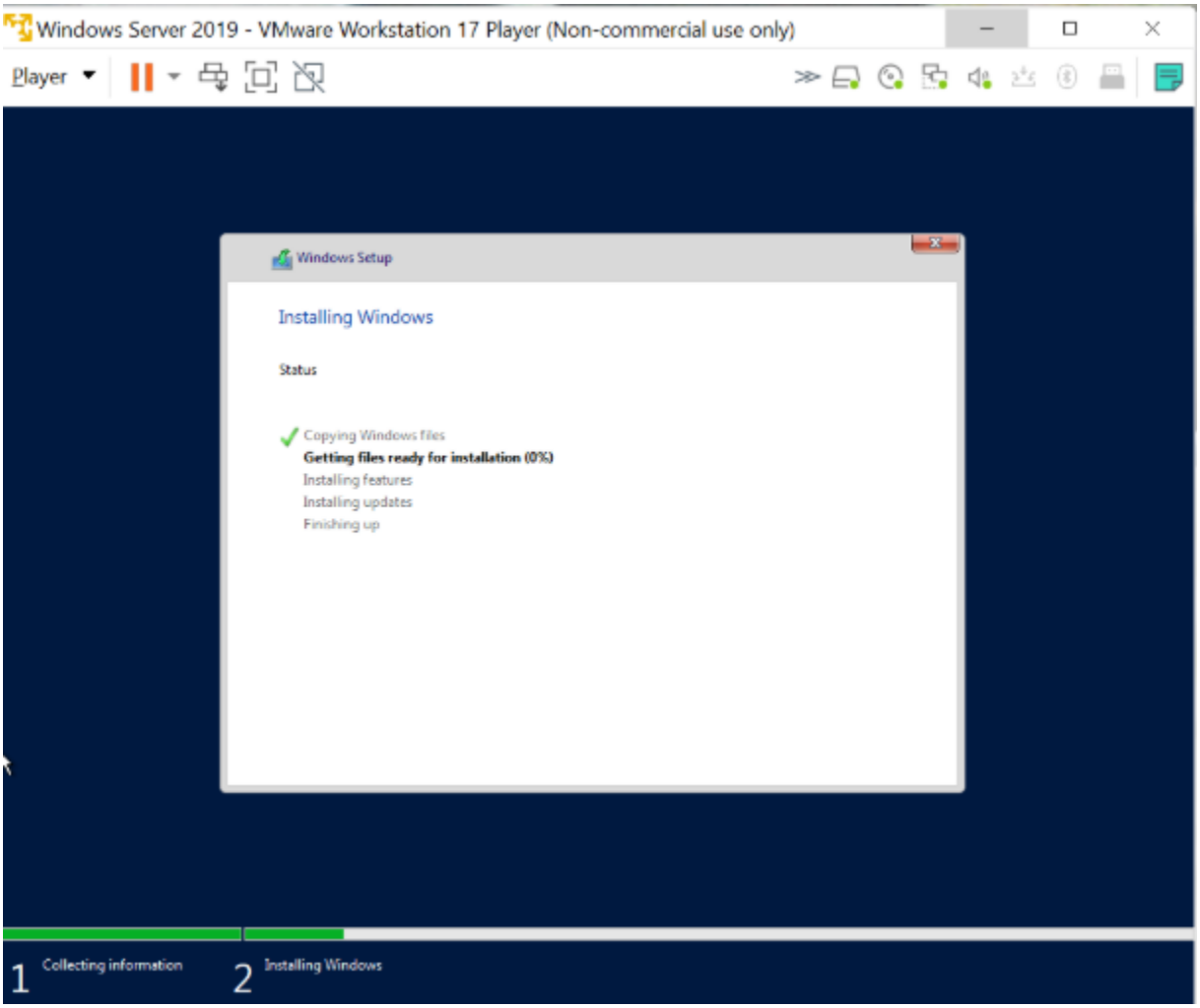


2. **Create a new VM** in VMware with at least:
    - **2GB RAM**, **2 CPU cores**, **40GB disk** (Minimum requirement for the windows 2019 server vm).
    - Attach the Windows Server ISO and complete the installation as like of opnsense.
    - **Note**: Use windows server standard edition as it require less resources and choose datacenter edition if your have large enterprise requirement.

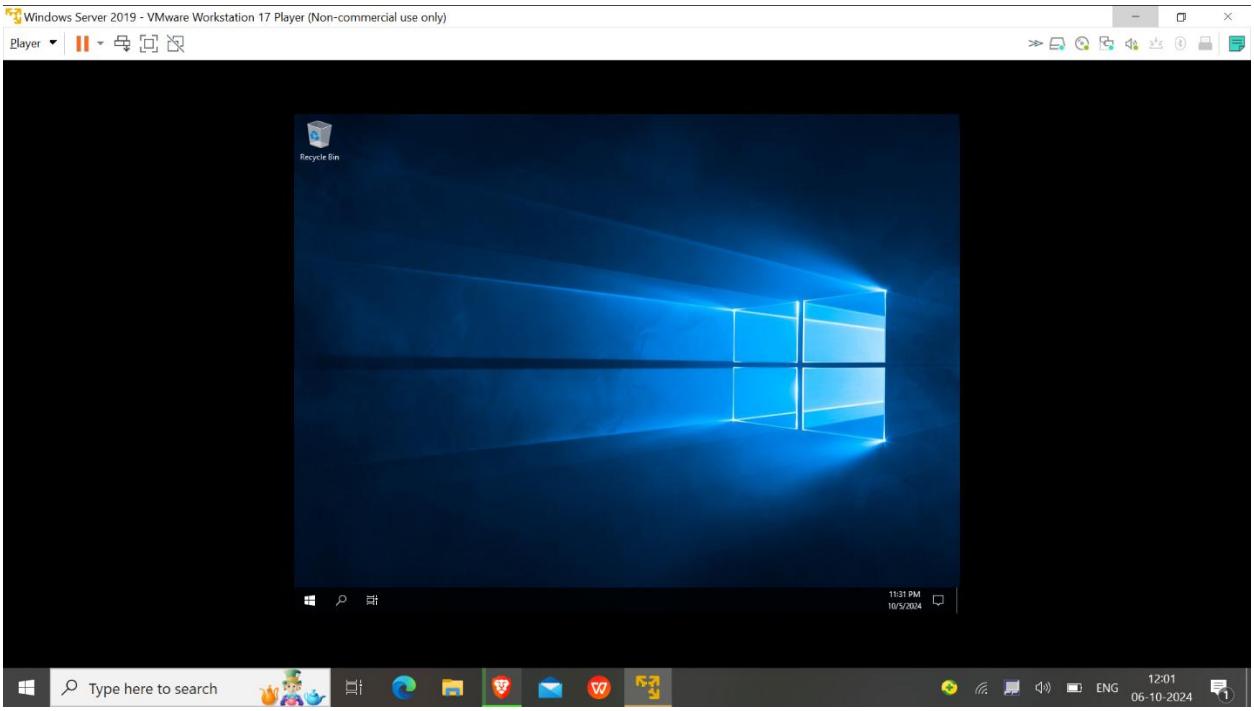



Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
2 GB
Processors
2
Hard Disk (NVMe)
40 GB
CD/DVD (SATA)
Using file D:\foldeerrr\17763.3...
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Display
Auto detect
Add...
Remove
Device status
Connected
Connect at power on
Connection
Use physical drive:
Auto detect
Use ISO image file:
D:\foldeerrr\17763.3650.221105-1748.rs5_release_
Browse...
Advanced...
OK
Cancel
Help

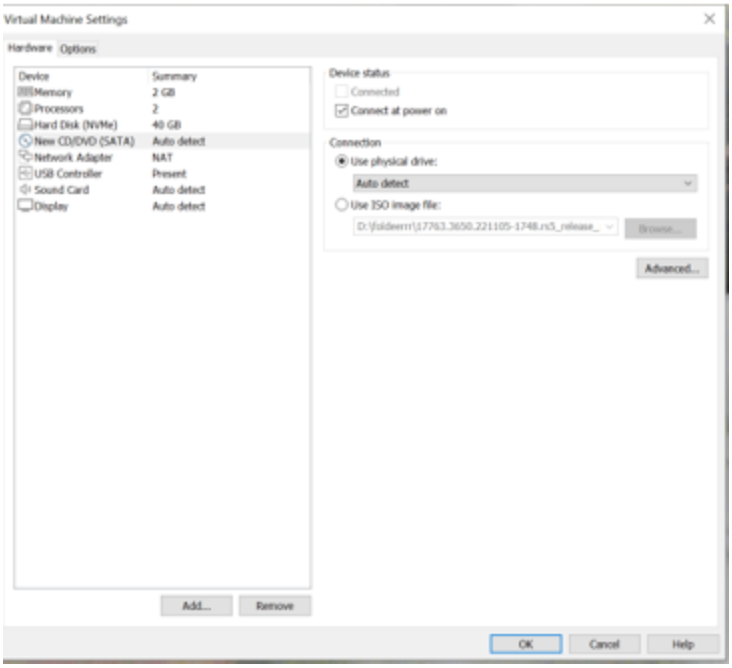

Windows Server 2019 - VMware Workstation 17 Player (Non-commercial use only)
Player
Windows Setup
Windows Server 2019
Language to install: English (United States)
Time and currency format: English (United States)
Keyboard or input method: US
Enter your language and other preferences and click "Next" to continue.
© 2018 Microsoft Corporation. All rights reserved.
Next



3. Set an **Administrator password** and log into the Windows Server.

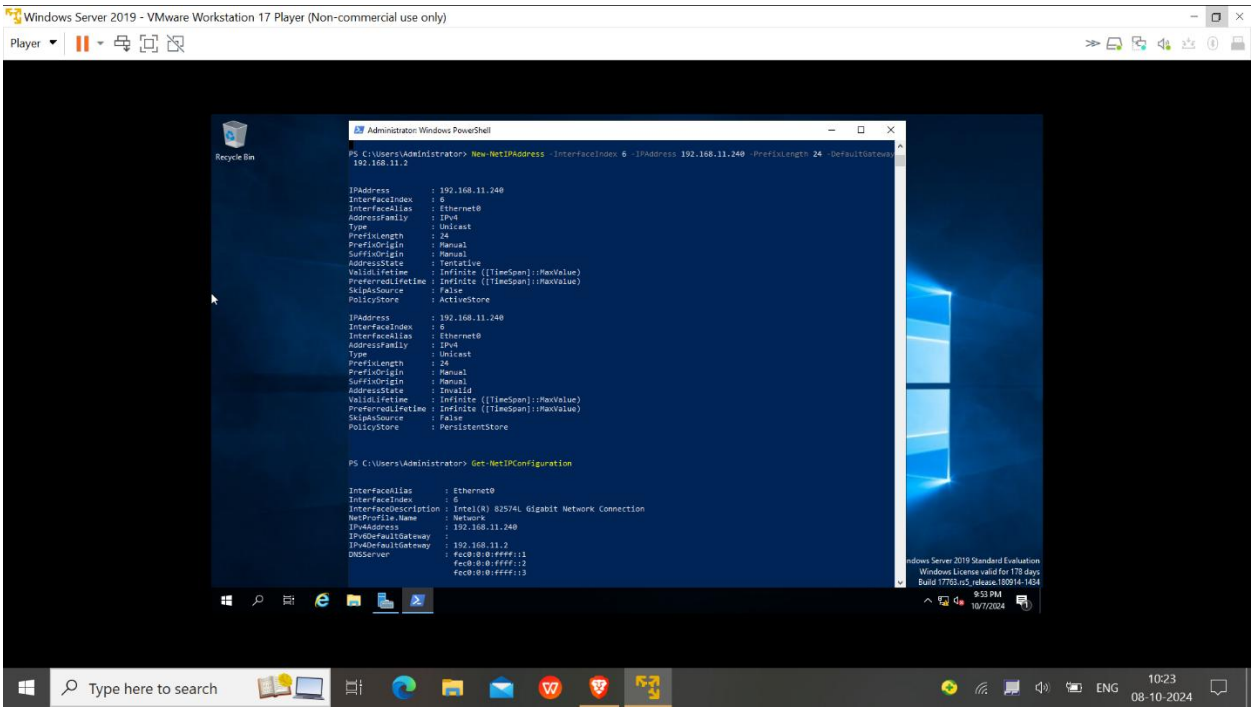


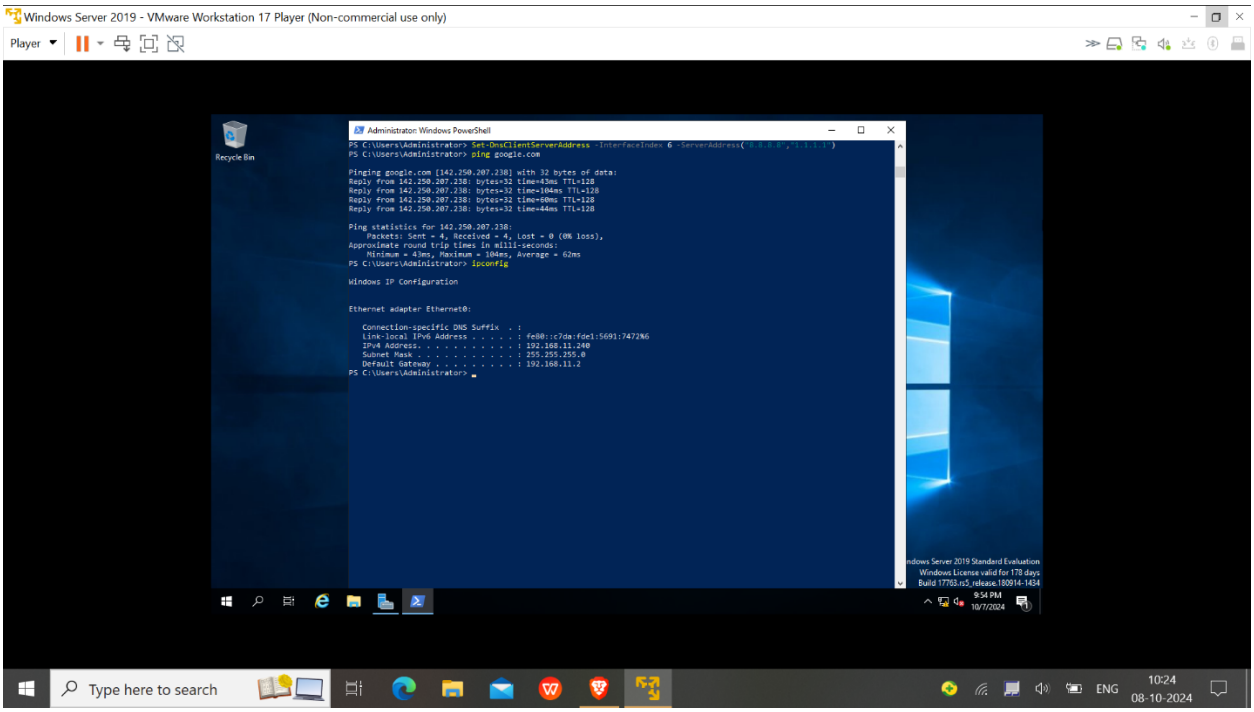
4. Remove the ISO from the booting device and set to directly boot from Hard disk.



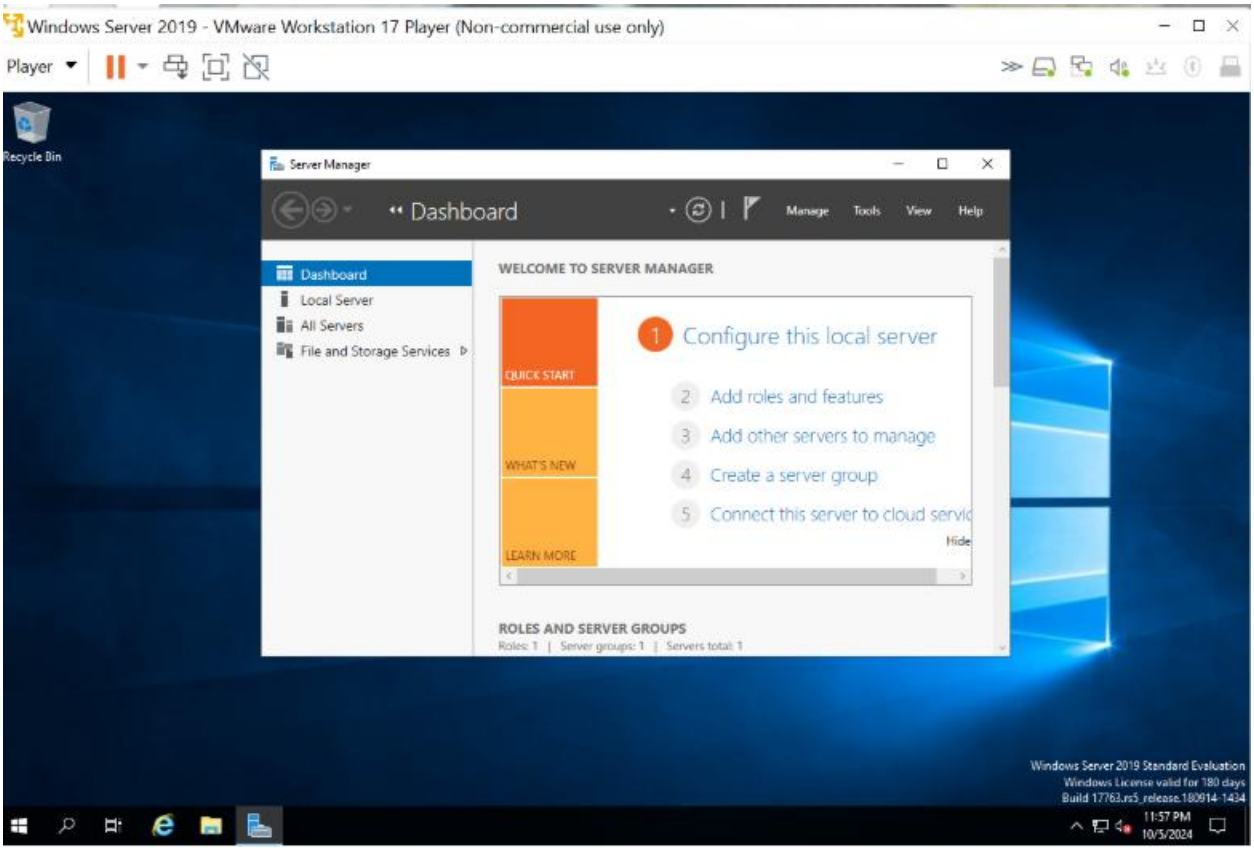
1.2 Configure Active Directory

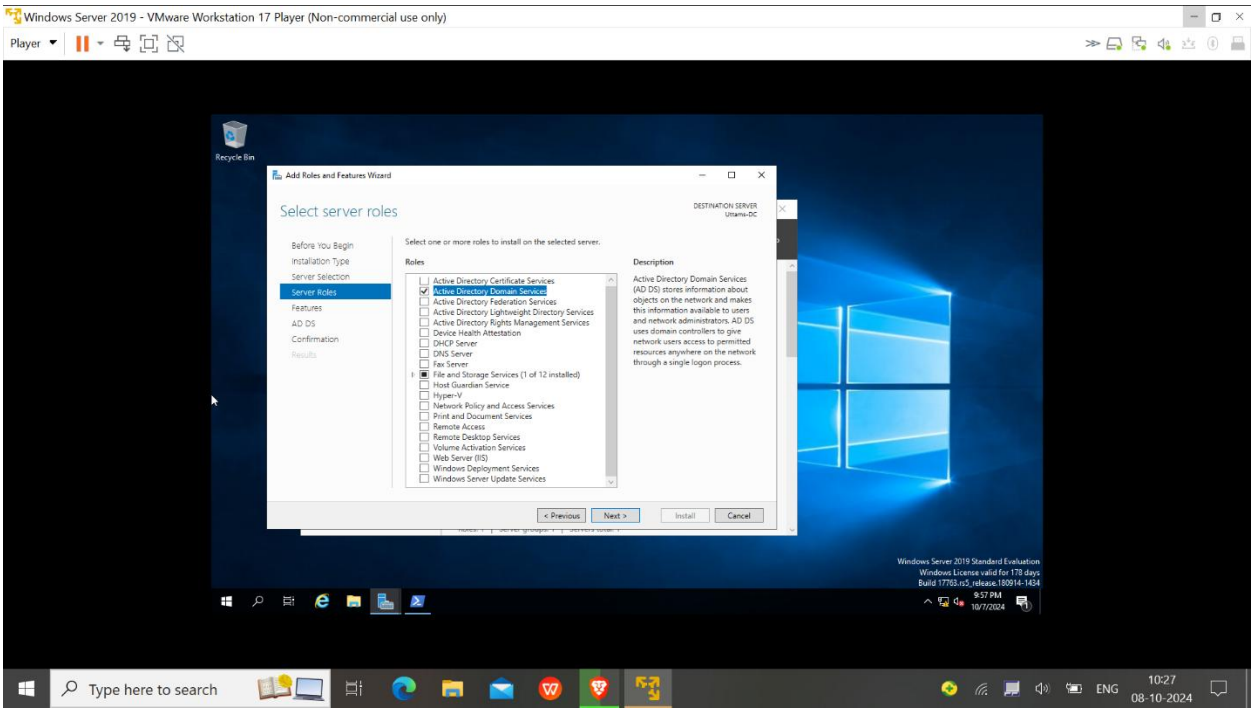
1. **Assign a Static IP along with subnet mask,default gateway and dns server** (In my case I use, ip =192.168.11.240 , subnet mask= 255.255.255.0, gateway= 192.168.11.2 and dns server =192.168.11.240).





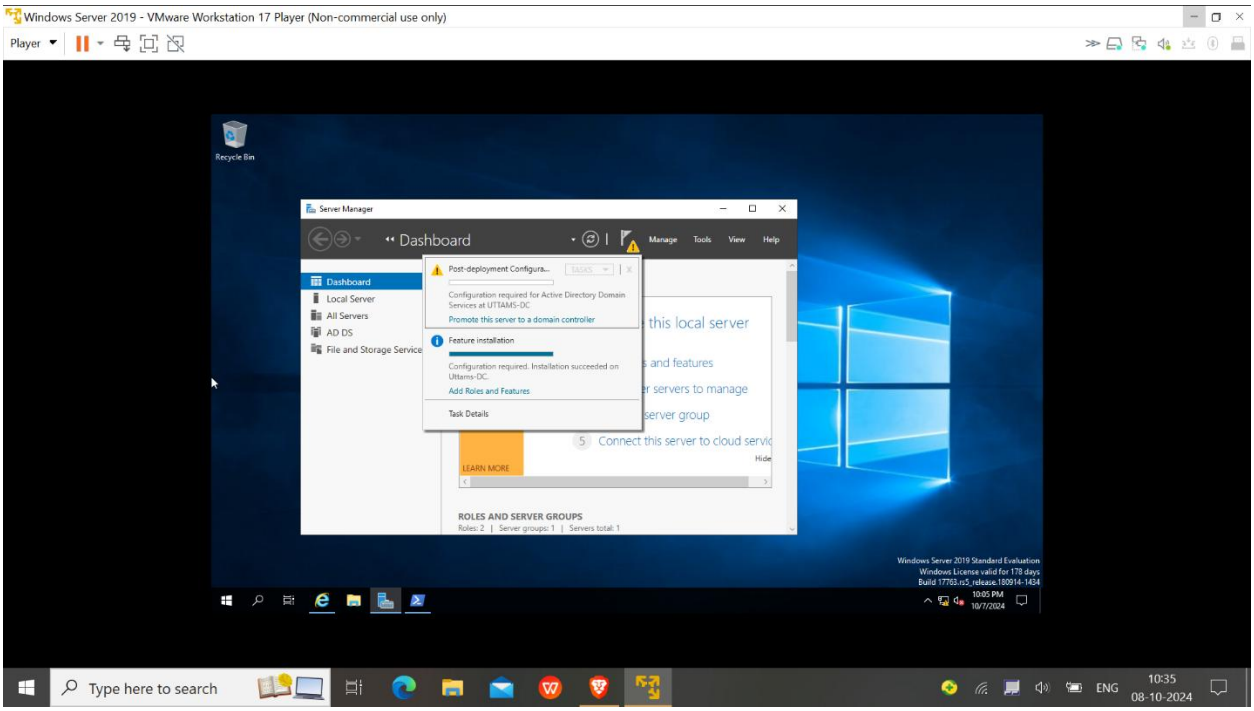
2. **Install Active Directory Domain Services (AD DS)** via **Server Manager > Add role and features**.

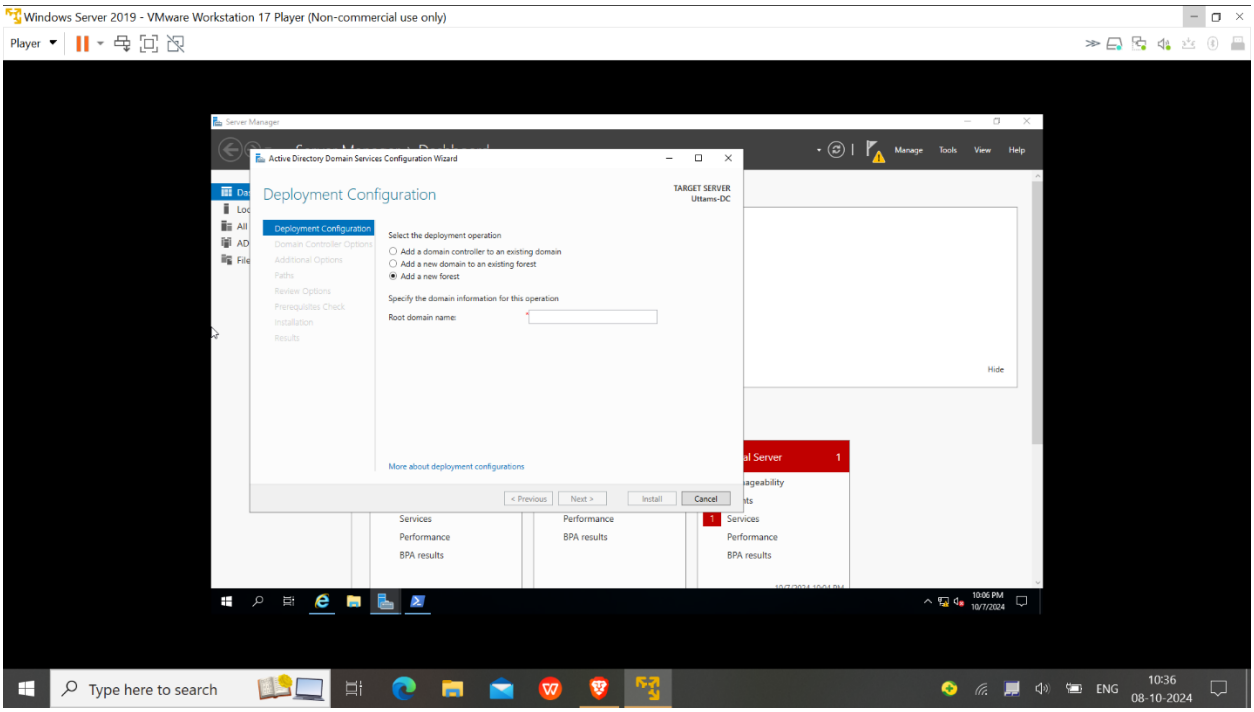




3. **Promote the server to a domain controller**:

    - Set up a new root domain (In my case I use, uttam.local).


Windows Server 2019 - VMware Workstation 17 Player (Non-commercial use only)
Player
Recycle Bin
Server Manager
Dashboard
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
AD DS
File and Storage Services
Post-deployment Configura...
Configuration required for Active Directory Domain Services at UTTAMS-DC
Promote this server to a domain controller
Feature installation
Configuration required. Installation succeeded on Uttams-DC.
Add Roles and Features
Task Details
this local server
and features
servers to manage
server group
Connect this server to cloud servic
LEARN MORE
Hide
ROLES AND SERVER GROUPS
Roles: 2 | Server groups: 1 | Servers total: 1
Windows Server 2019 Standard Evaluation
Windows License valid for 178 days
Build 17763.rs5_release.180914-1434
10:05 PM
10/7/2024
Type here to search
ENG
10:35
08-10-2024


Windows Server 2019 - VMware Workstation 17 Player (Non-commercial use only)
Player
Server Manager
Manage
Tools
View
Help
Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
Uttams-DC
Deployment Configuration
Domain Controller Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
More about deployment configurations
< Previous
Next >
Install
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Services
Performance
BPA results
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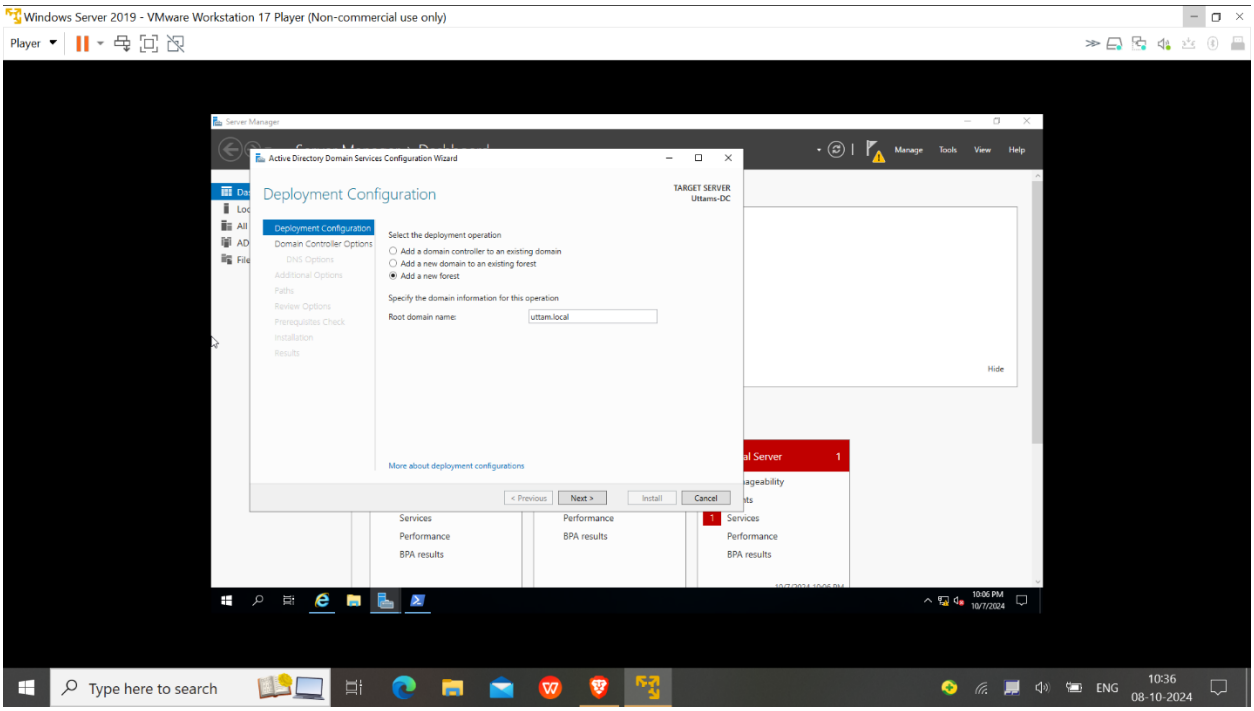
Windows Server 2019 - VMware Workstation 17 Player (Non-commercial use only)
Player
Server Manager
Manage
Tools
View
Help
Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
Uttams-DC
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
uttam.local
More about deployment configurations
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Install
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Services
Performance
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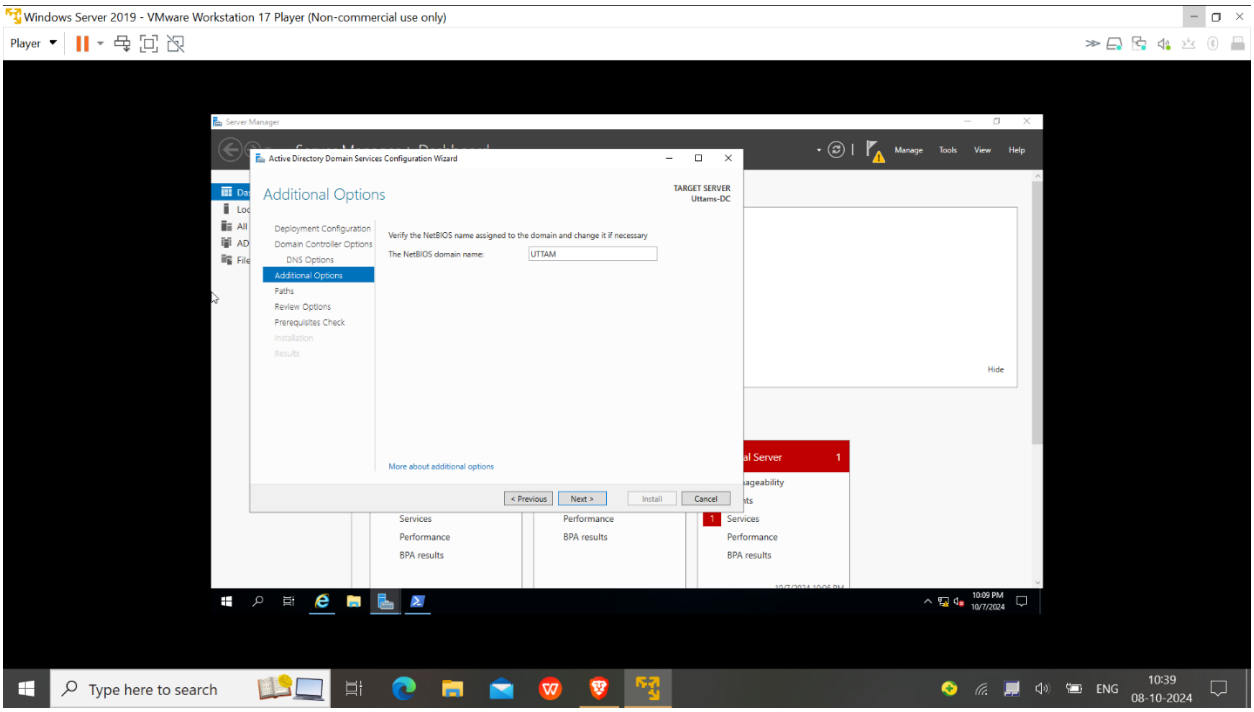
Windows Server 2019 - VMware Workstation 17 Player (Non-commercial use only)
Player
Server Manager
Manage
Tools
View
Help
Active Directory Domain Services Configuration Wizard
Additional Options
TARGET SERVER
Uttams-DC
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Verify the NetBIOS name assigned to the domain and change it if necessary
The NetBIOS domain name:
UTTAM
More about additional options
< Previous
Next >
Install
Cancel
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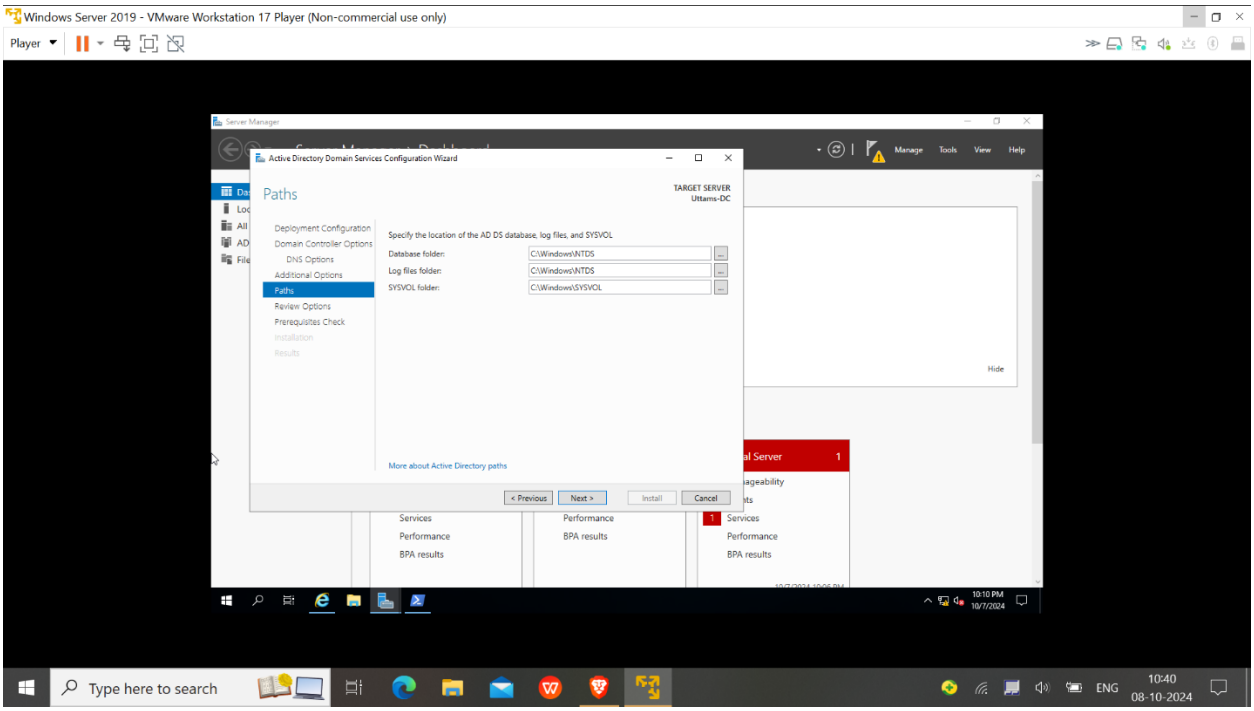
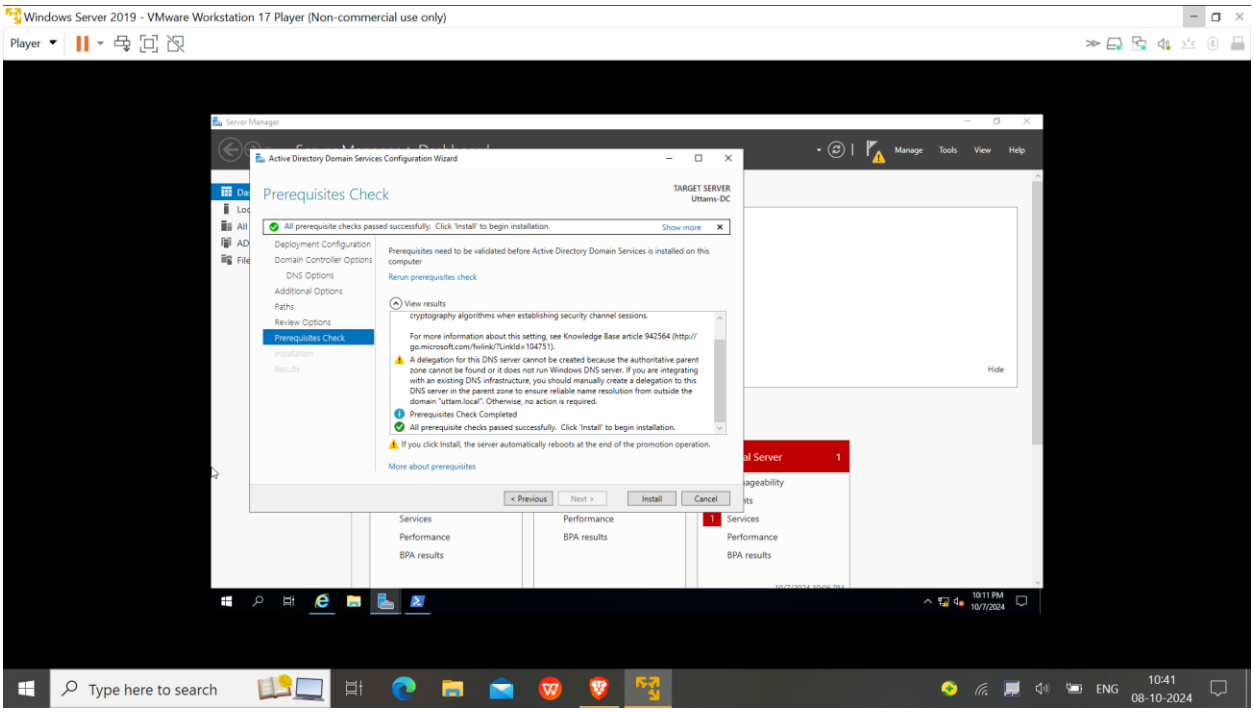
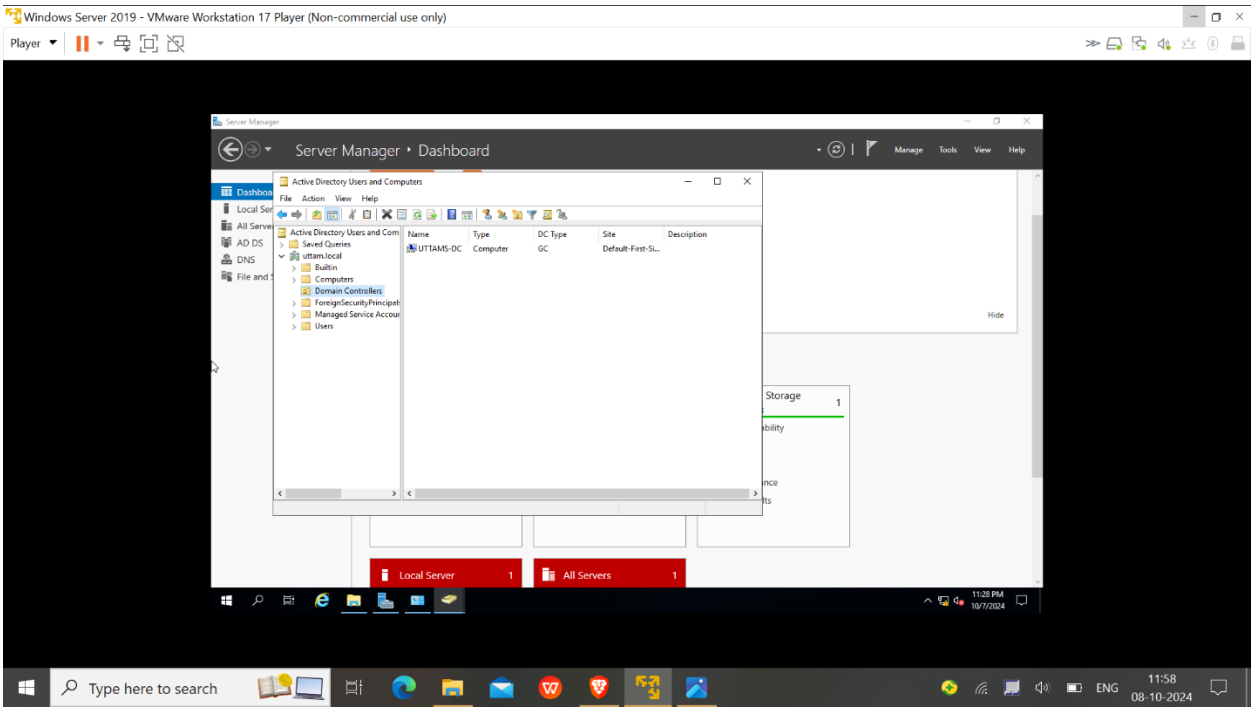
Services
Performance
BPA results
Type here to search
ENG
10:39
08-10-2024





4. Now Restart the server and open **Server Manager> Tools> Active Directory Users and Computers** to verify that AD is running.

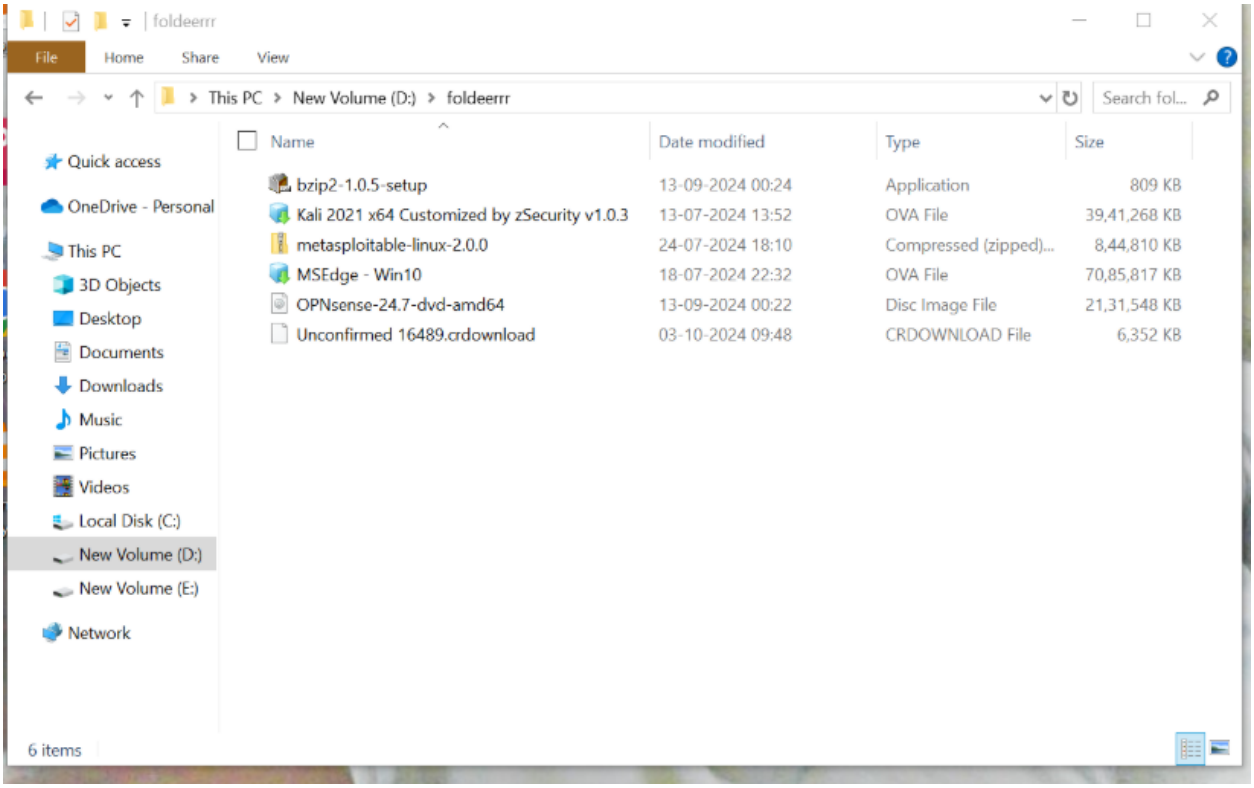


Step 2: Set Up Windows 10 Client VM

The Windows 10 client will be joined to the domain controlled by the Windows Server.

2.1 Install Windows 10 on VMWare

1. **Download Windows 10 OVA file** from Microsoft's website.

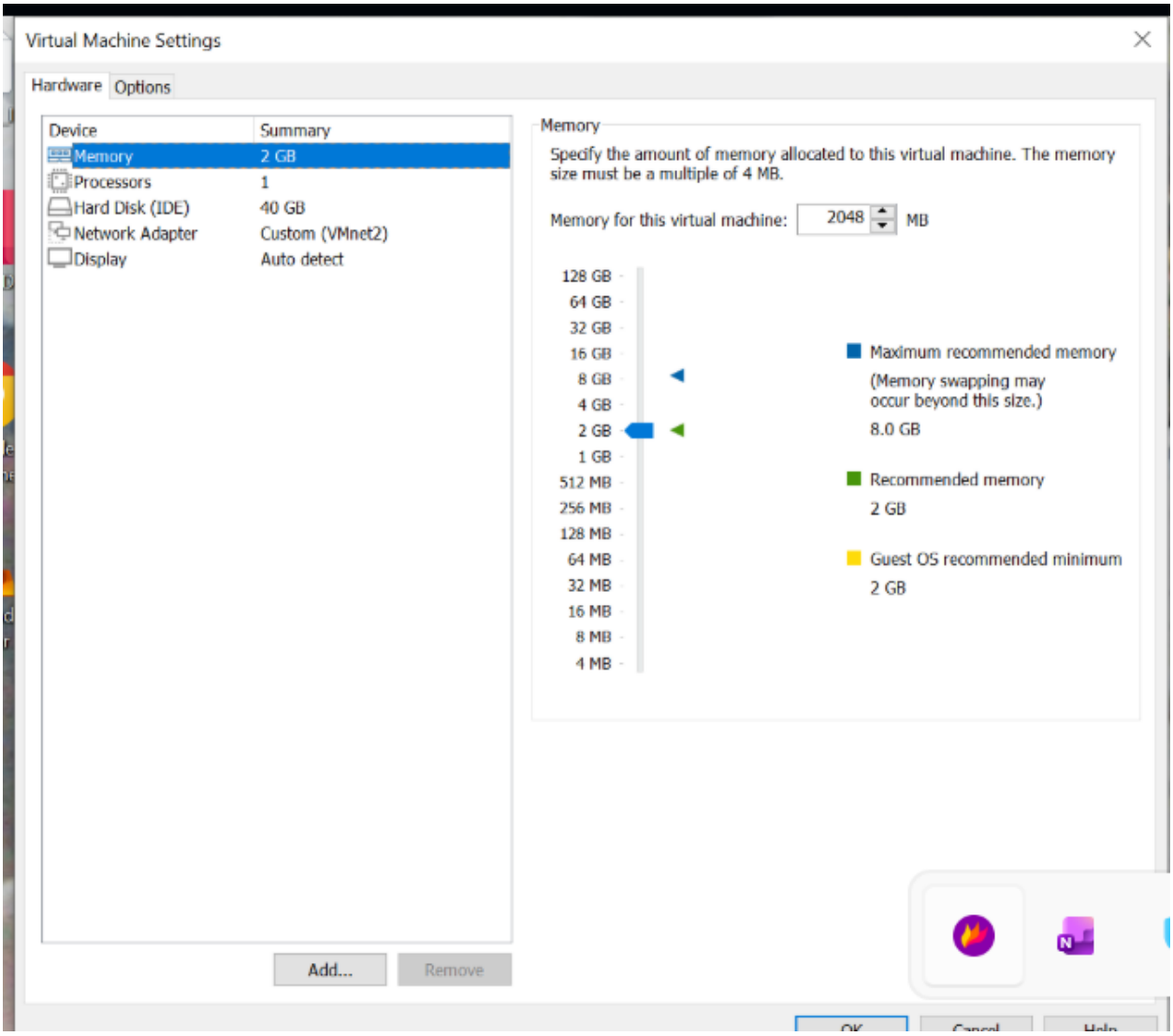


2. From **OVA package** you had a Pre-configured VM which can be imported directly and ready to use without running the intaller. **Alternative : from ISO image**

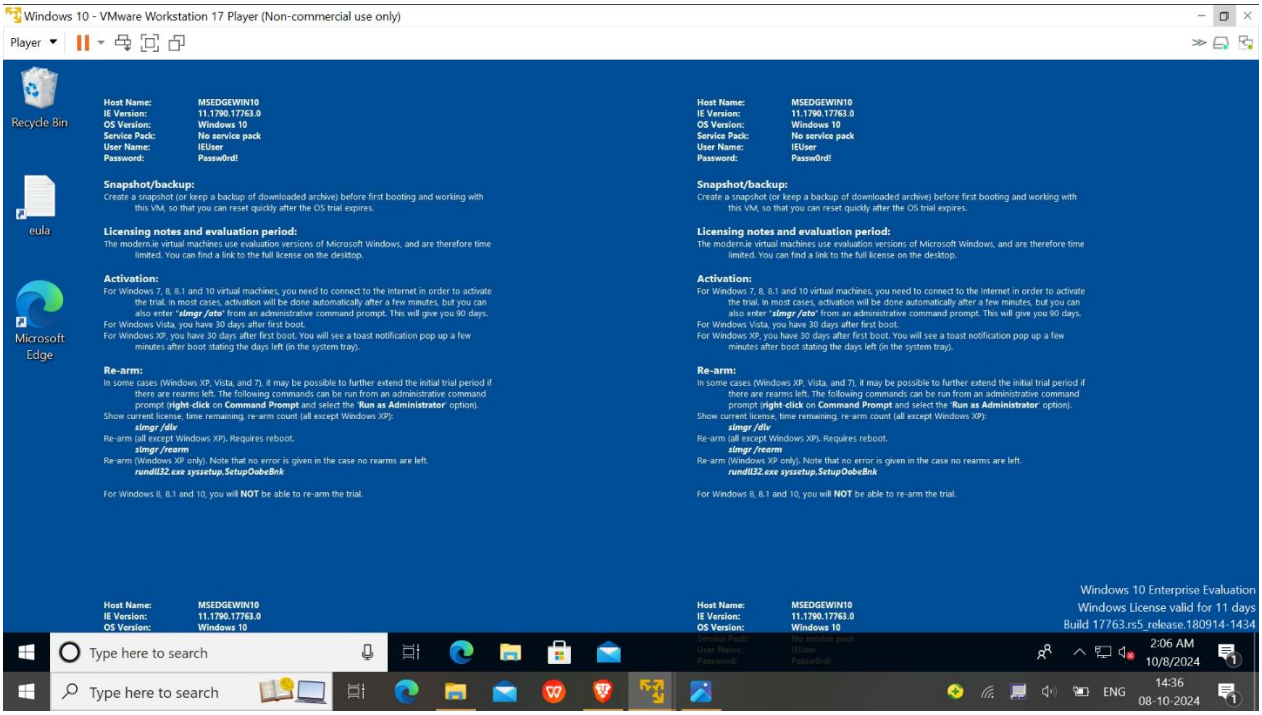
3. **Download Windows 10 ISO file** from Microsoft's website.

4. **Create a new VM** in VMware with:

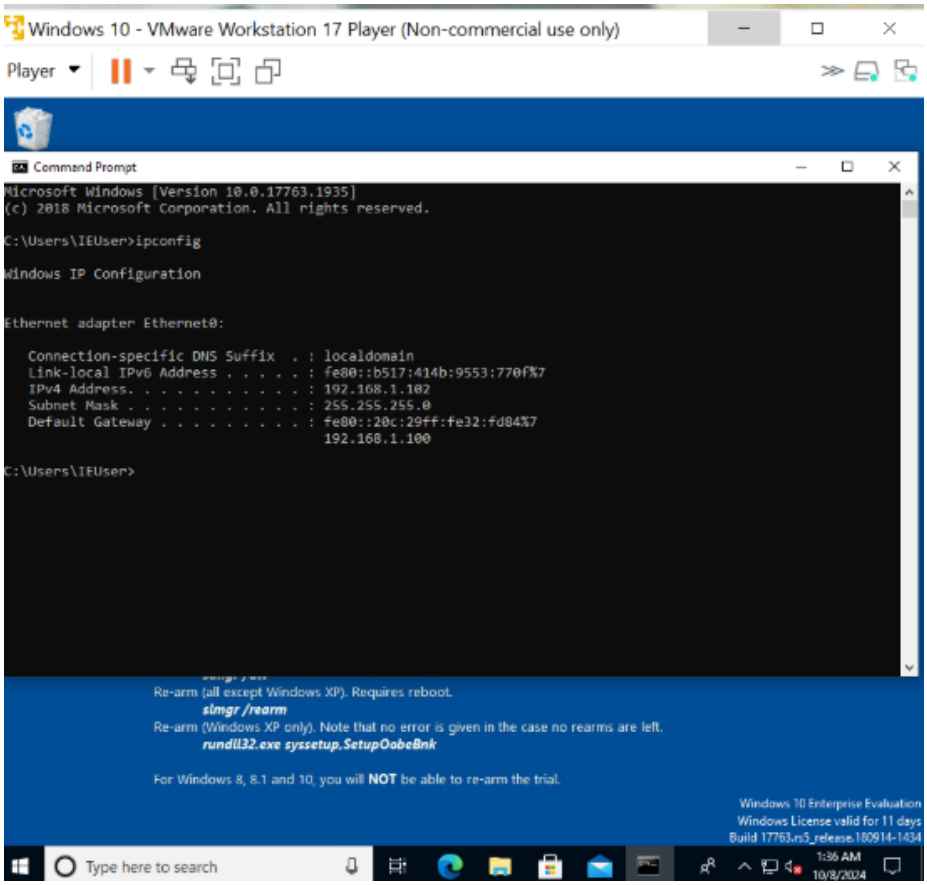
    - **1.5GB RAM, 1 CPU core, 40GB disk.**



5. Complete the installation with a local account similar to earlier ISO installation and start the machine.

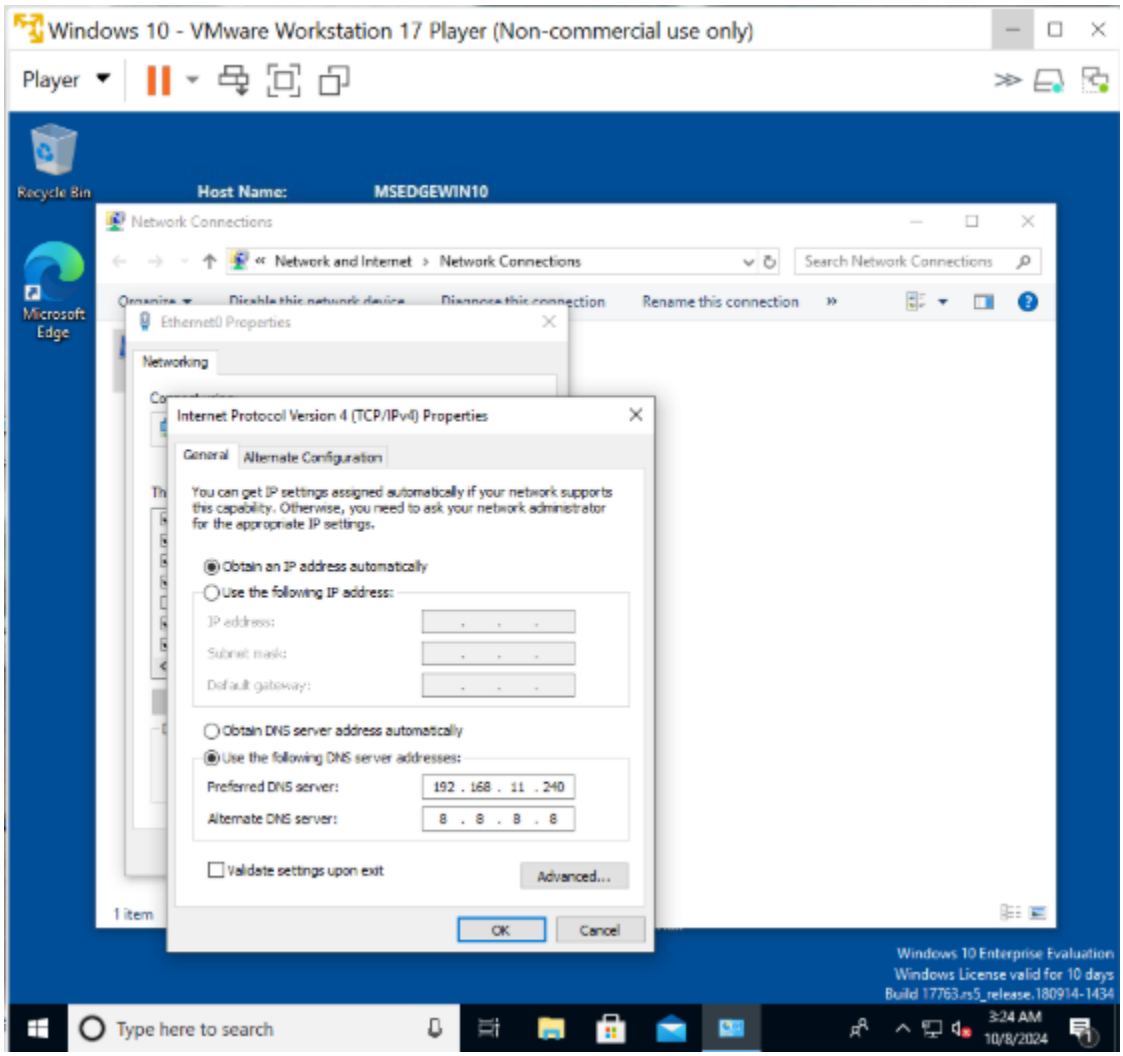


6. **Check network configuration**



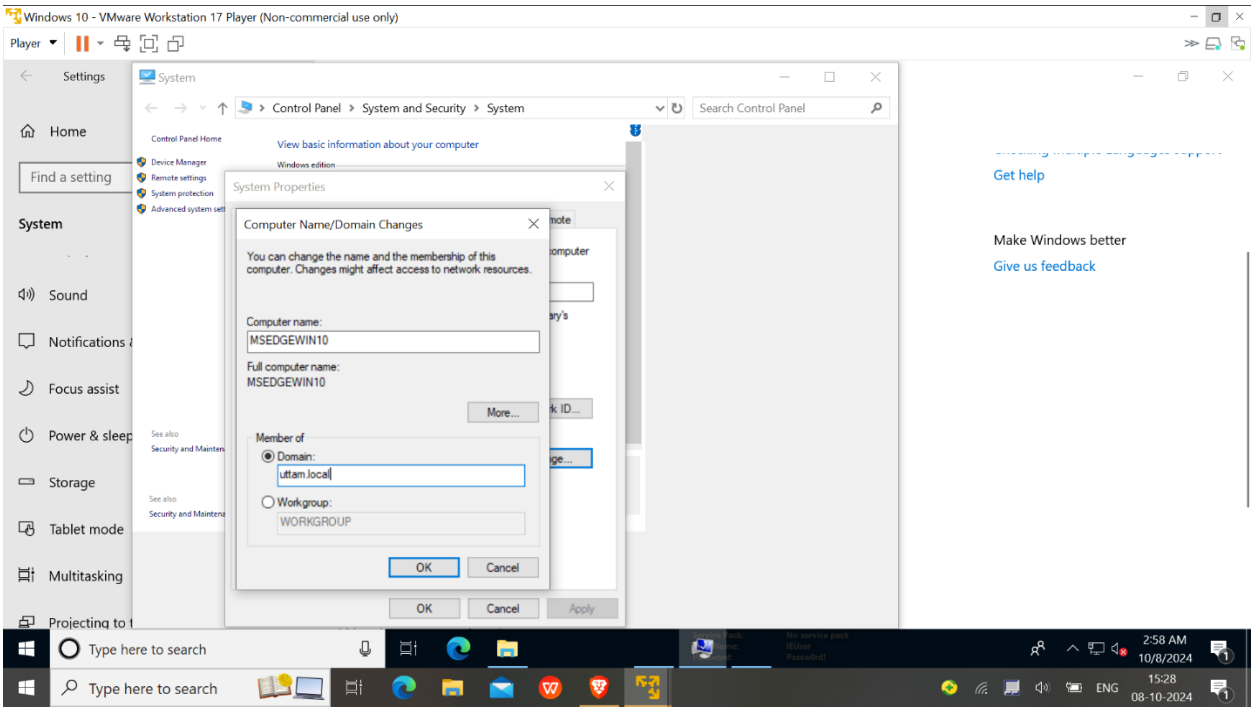
2.2 Join the Windows 10 Client to the Domain

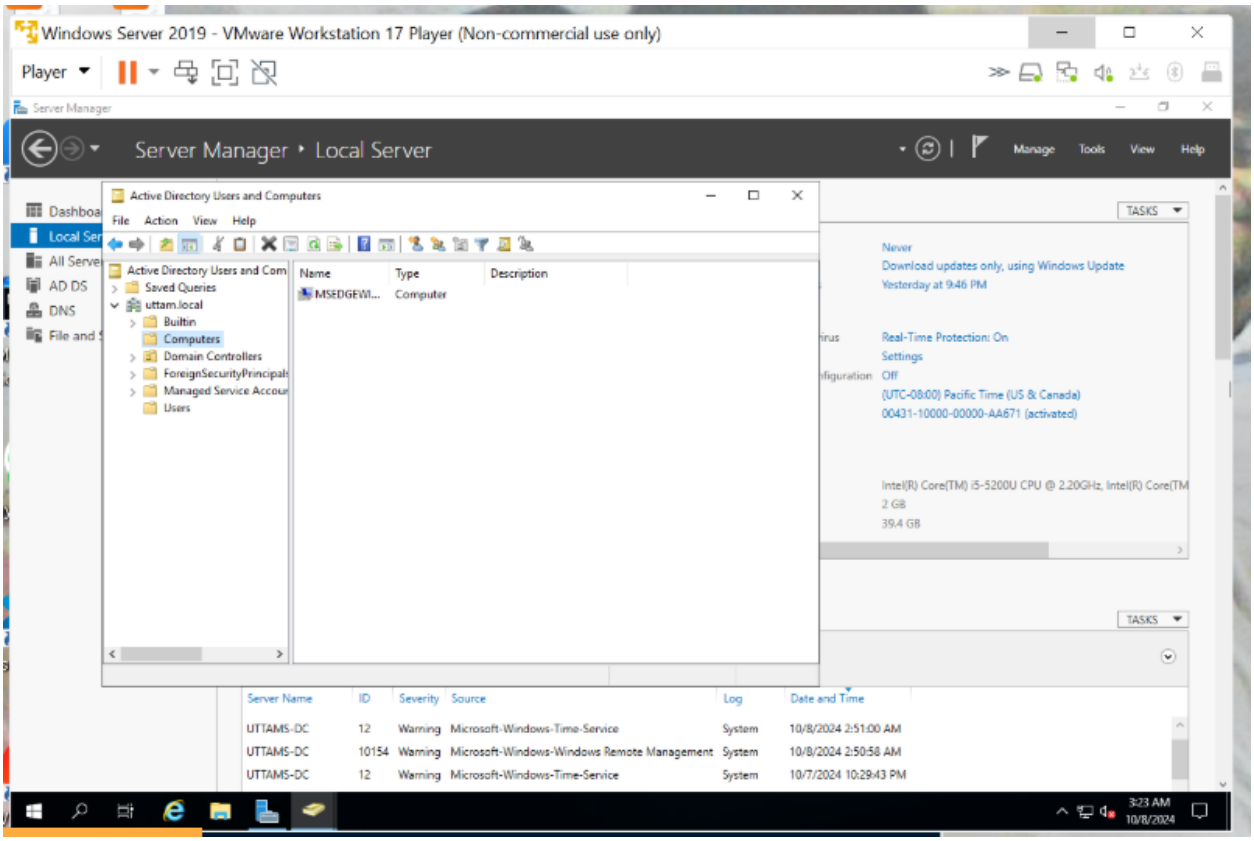
1. Set the **DNS server** of the Windows 10 VM to the IP of the Windows Server (In my case, 192.168.11.240).



2. Go to **System > About** and click **Advance Settings**.

3. Enter the **domain name** (In my case, uttam.local), and provide **Administrator credentials or any user crudential that is present on the DC** to join the domain.





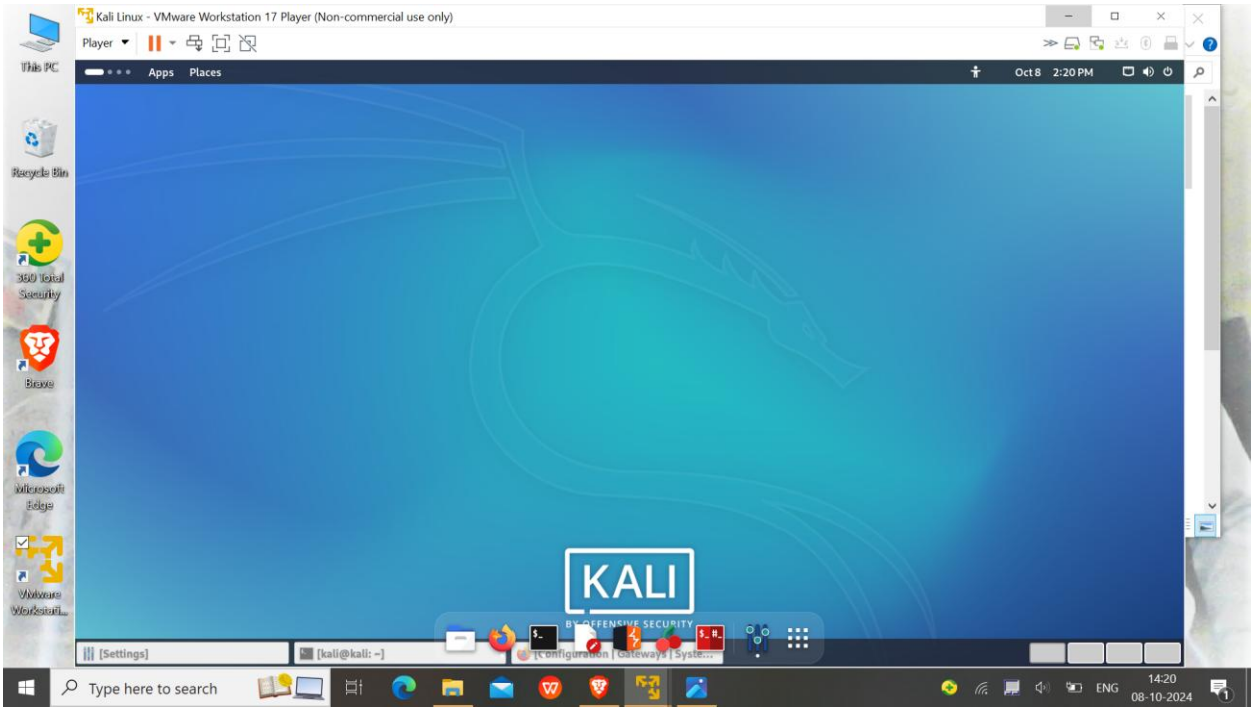
4. Restart the VM and log in with **domain credentials**.

Step 3: Set Up Kali Linux for Penetration Testing

Kali Linux will be used for penetration testing(attacking machine) in my homelab. It will be used to test attacks against both the **Windows Server, windows 10 client** and **Metasploitable 2**.

3.1 Install Kali Linux on VM

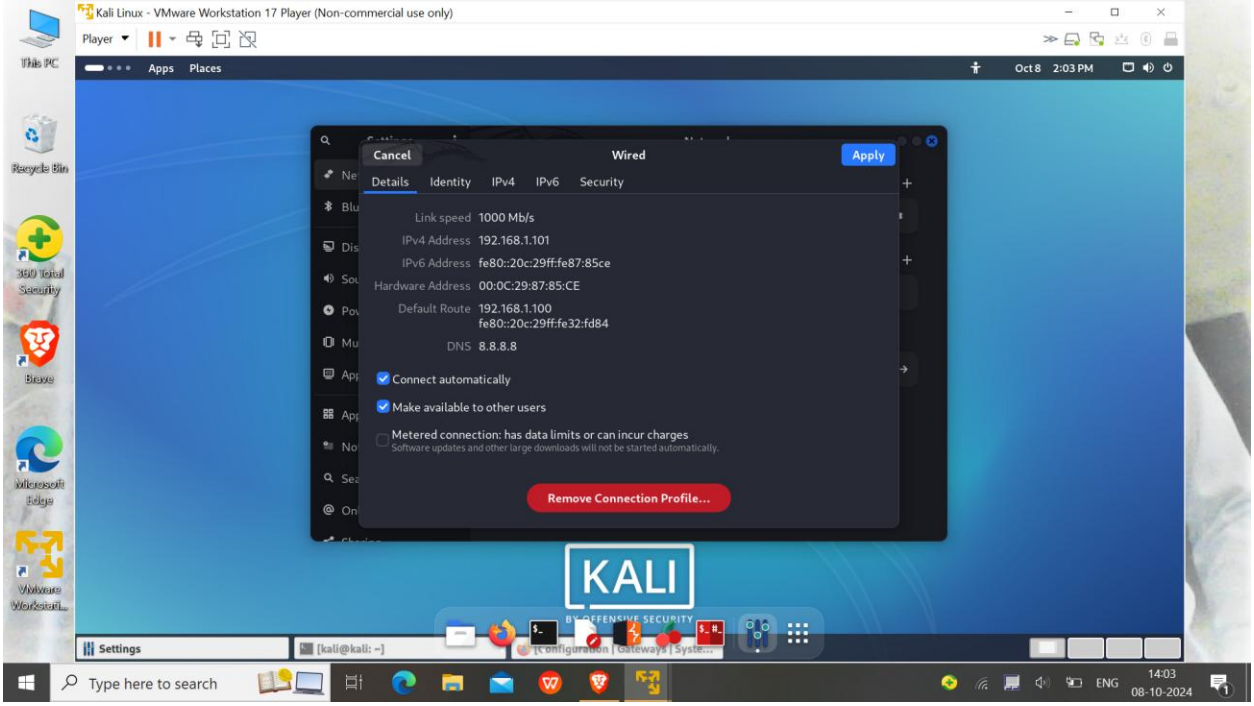
- You can install kali from the **ISO** or through a pre-configured **OVA** file(Similar installation steps).
- Both **ISO** and **OVA** files are available at kali linux official website.
- Once installed, update the system by running following command in terminal:
- sudo apt-get update && sudo apt-get upgrade



3.2 Configure Kali Networking

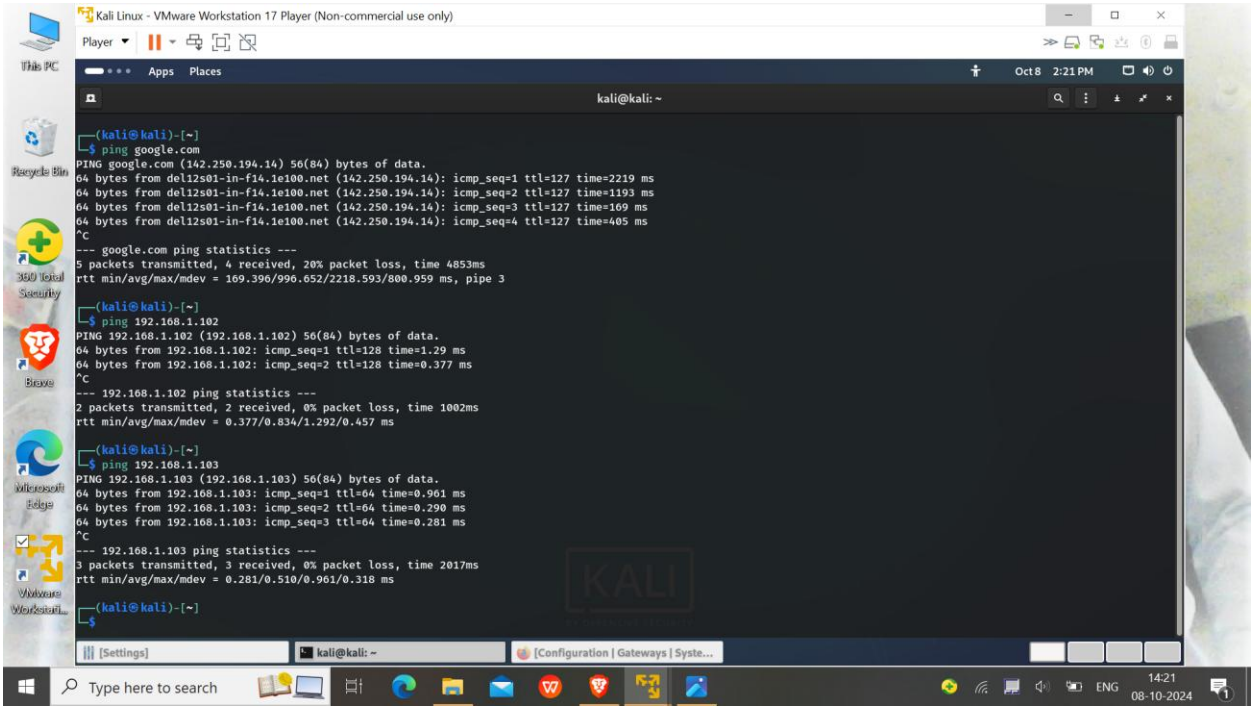
Configure the Network:

1. Set **Kali Linux** to use the same **internal network (vmnet 2)** as the other VMs in the homelab (LAN network of OPNsense).
2. The default gateway is set to the **LAN ip of opnsense**.(In my case it is 192.168.1.100)



Verify Connectivity:

Check if the Kali Linux VM can communicate with the other VMs in your network: bash ping 192.168.11.240 # Ping Windows Server ping 192.168.1.102 # Ping Windows 10 Client ping 192.168.1.103 # Ping Metasploitable 2 ping google.com # Ping to internet

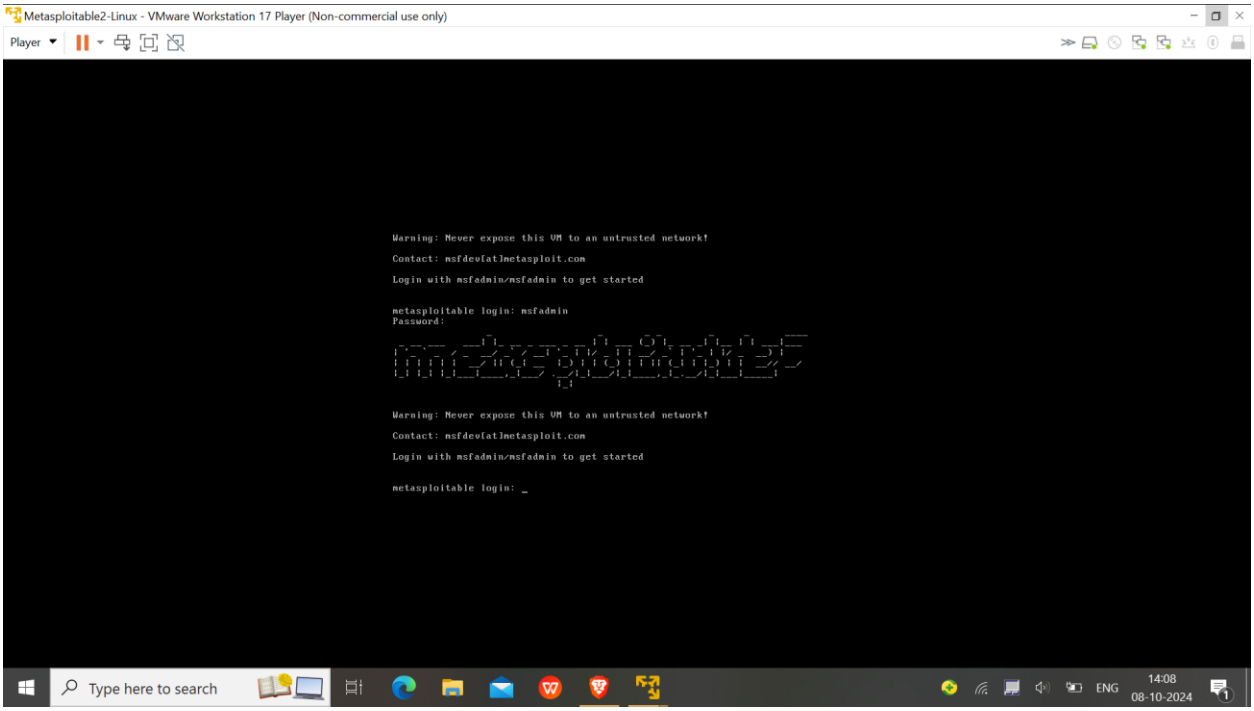


Step 4: Set Up Metasploitable 2 (Vulnerable Target Machine)

Metasploitable 2 is a deliberately vulnerable Linux machine designed to be used for practicing penetration testing.

4.1 Install Metasploitable 2

1. **Download Metasploitable 2** from official website: https://sourceforge.net/projects/metasploitable/.
2. **Import the VM into VMware**:
   - Open VMware and select **Import Appliance**.
   - Import the **Metasploitable 2 OVA** file.
3. **Start the VM** and log in with default crudential **user-> msfadmin and password-> msfadmin**.

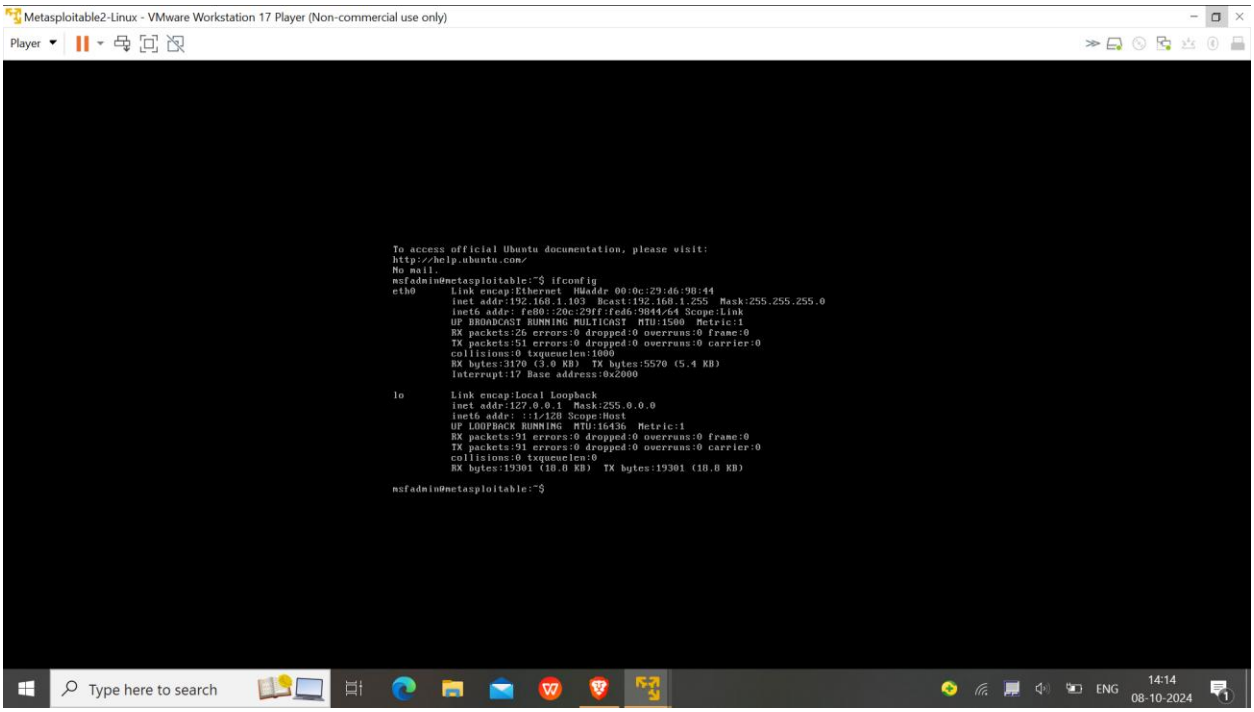


4.2 Configure Networking for Metasploitable 2

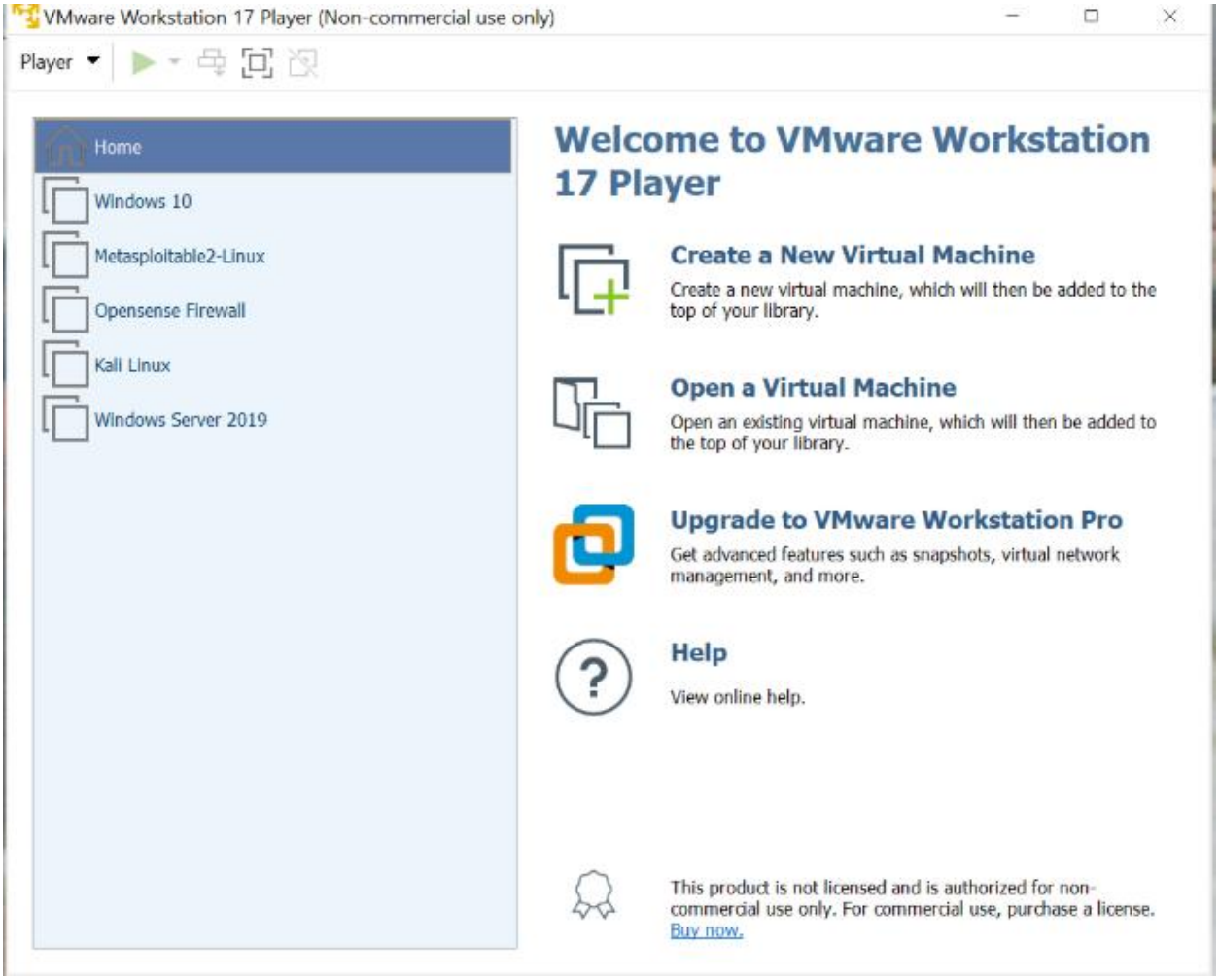
1. Set the **network adapter** of Metasploitable 2 to the same internal network (vmnet 2) as the other VMs.

Verify Connectivity:

Ensure that **Metasploitable 2** can be pinged from **Kali Linux** and other VMs.



Now my Homelab is completely setup and let's dive into the various attack scenarios


VMware Workstation 17 Player (Non-commercial use only)
Player
Home
Windows 10
Metasploitable2-Linux
Opensense Firewall
Kali Linux
Windows Server 2019
Welcome to VMware Workstation 17 Player
Create a New Virtual Machine
Create a new virtual machine, which will then be added to the top of your library.
Open a Virtual Machine
Open an existing virtual machine, which will then be added to the top of your library.
Upgrade to VMware Workstation Pro
Get advanced features such as snapshots, virtual network management, and more.
Help
View online help.
This product is not licensed and is authorized for non-commercial use only. For commercial use, purchase a license.
Buy now.